American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

Evaluate and express the answer in hexadecimal.

$100_2 \bullet (42_8 + 375_8)$

A. 43700
B. 11F
C. 437
D. 47C
E. 2174

2. Computer Number Systems

In what year after 2025_{10} will the year, expressed in hexadecimal, first be a palindrome?

A. 707
B. 7E7
C. 7F7
D. 8E8
E. 9E9

3. Recursive Functions

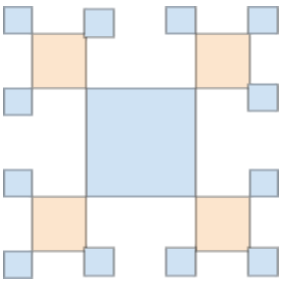
Find $f(f(f(7)))$ given:

$$f(x) = \begin{cases} f(x+3) - 1 & \text{if } x \text{ is a prime number greater than } 5 \\ f(x-1) + 2 & \text{if } x = 1 \text{ or } x \text{ is a multiple of either } 3 \text{ or } 5 \\ x - 2 & \text{otherwise} \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 10

4. Recursive Functions

Step 1: Draw a square.
Step 2: Draw new squares at each corner of the original square, each of which is half the length.
Repeat Step 2 four more times without using the connected corner.
How many squares were added during the final repetition?



A. 108
B. 243
C. 256
D. 324
E. 1024

5. What Does This Program Do?

What is the output of the following program if array *arr* is filled with the following values?

arr=	1000	500	100	50	10	5	1

```
letters = "MDCLXVI"
num = 1776
ans = ""
for x = 0 to 6
    val = int(num / arr(x))
    for y = 1 to val
        ans = ans + letters[x]
    next y
    num = num % arr(x)
next x
output ans
```

A. MDCCCLXVI
B. MDCCLXXVI
C. MDCLXXVII
D. MDCCXXXVI
E. DDCCLXXVI

6. What Does This Program Do?

What is the output of the following program if the function str(*n*) makes the number *n* a string value?

```
n = 0
for x = 1 to 4
    z = 10 - x
    a = str(1089 * x)
    s = ""
    for y = 0 to len(a)-1
        s = a[y] + s
    next y
    if s == str(1089 * z)
        n = n + 1
    end if
next x
output n
```

A. 4
B. 3
C. 2
D. 1
E. 0

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2025-2026 ● Contest 1: Short Problems ● Senior Division

1. 计算机数字系统

求值并以十六进制表示结果。

$$100_2 \bullet (42_8 + 375_8)$$

A. 43700
B. 11F
C. 437
D. 47C
E. 2174

2. 计算机数字系统

以 16 进制表示 2025_{10} 之后的年份，哪一年将首次以回文数表示？

A. 707
B. 7E7
C. 7F7
D. 8E8
E. 9E9

3. 递归函数

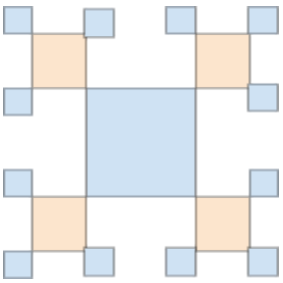
根据以下表达式，求 $f(f(f(7)))$ ：

$$f(x) = \begin{cases} f(x+3) - 1 & \text{如果 x 是大于 5 的质数} \\ f(x-1) + 2 & \text{如果 x=1，或 x 是 3 或 5 的倍数} \\ x - 2 & \text{否则} \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 10

4. 递归函数

步骤 1：画一个正方形
步骤 2：在原正方形的每个顶点处，各画一个新正方形。每个新正方形的边长为原正方形的一半。
在不使用已连接的顶点的情况下，再重复步骤 2 四次。



最后一次重复步骤时增加了多少个正方形？

A. 108
B. 243
C. 256
D. 324
E. 1024

5. 这个程序是做什么的？

若数组 *arr* 中填入以下数值，该程序的输出结果是什么？

arr=	1000	500	100	50	10	5	1

```
letters = "MDCLXVI"
num = 1776
ans = ""
for x = 0 to 6
    val = int(num / arr(x))
    for y = 1 to val
        ans = ans + letters[x]
    next y
    num = num % arr(x)
next x
output ans
```

A. MDCCCLXVI
B. MDCCLXXVI
C. MDCLXXVII
D. MDCCXXXVI
E. DDCCLXXVI

6. 这个程序是做什么的？

如果函数 str(*n*) 将数字 *n* 转换为字符串值，以下程序的输出是什么？

```
n = 0
for x = 1 to 4
    z = 10 - x
    a = str(1089 * x)
    s = ""
    for y = 0 to len(a)-1
        s = a[y] + s
    next y
    if s == str(1089 * z)
        n = n + 1
    end if
next x
output n
```

A. 4
B. 3
C. 2
D. 1
E. 0